TOPIC 3 · CONSERVATION LAWS

Solved Problems: Official Collection CL1 to CL5

Worked solutions to the conservation-laws collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid (5 of the 22 problems so far; the rest of the collection is added in batches). Every problem restates the official statement, then follows four phases: (1) hypotheses, data and unknowns; (2) frames, control volume and its surfaces; (3) the balances; (4) interpretation, limits and dimensional check. The balances follow the method used in class: each one is written in full with the vanishing terms crossed out and the reason underneath, every surface integral is then evaluated surface by surface (the vertical line drawn under each integral), the wall terms are grouped (boxed) as the force on the body, and each result is boxed and numbered. Discrepancies found in the handwritten official solutions are flagged in the corresponding problem, and anything beyond the course material is marked as an extension. References to Notes.pdf are to Chapter 3 (Conservation laws).

PROBLEM CL1

Airfoil Wake: Streamline Displacement and Drag

Original statement · CL1.pdf

Control volume | Linear momentum | Wake | Drag

OFFICIAL PROBLEM STATEMENT

The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure of CL1.pdf. For the airfoil tested, the velocity measurements at a given downstream section reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \to -\infty$, $y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0$, $y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0, as well as the value of the drag on the airfoil as a function of ρ, y_0 and U. In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_∞ are negligible, and so is the effect of gravity on the air flow.

Source: CL1.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36, force of the fluid on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (everything per unit span), incompressible (ρ constant); neglect $\bar{g}$.
- Viscous stresses negligible in the wake and on the bounding streamlines; $p - p_\infty \simeq 0$ on the whole outer boundary of the control volume (given). On the airfoil surface pressure and viscous stresses are **not** neglected: they make the drag.
- Wake profile at $x = x_0$ (symmetric about $y = 0$): $v_x = \frac{U}{2}\left(1 + \frac{|y|}{y_0}\right)$ for $|y| \leq y_0$, and $v_x = U$ outside.
- Data: $[\rho,\ U,\ y_0]$. Unknowns: $[y_{-\infty},\ D]$, with D the drag per unit span.

Symbol	Meaning	SI unit
U	uniform upstream velocity	m/s
y_0	half-width of the wake at $x = x_0$	m
$y_{-\infty}$	half-width of the stream tube far upstream	m
ρ	air density	kg/m^3
D	drag per unit span	N/m

Dimensional check of the unknown: $[\rho U^2 y_0] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m} = \mathrm{N/m}$, a force per unit length, as required for a planar problem.

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the airfoil and the tunnel; basis $(\bar{e}_x, \bar{e}_y, \bar{e}_z)$ with $\bar{e}_x$ along the stream. The control volume is fixed: $\bar{v}_c = 0$.
- Surfaces of Σ_c (outward normal $\bar{n}$): Σ_1, far upstream, $|y| < y_{-\infty}$, $\bar{n} = -\bar{e}_x$; Σ_2, the section $x = x_0$, $|y| < y_0$, $\bar{n} = +\bar{e}_x$; Σ_l, the two bounding streamlines ($\bar{v} \cdot \bar{n} = 0$); Σ_w, the airfoil surface ($\bar{n}$ pointing into the airfoil).
- Change of variable for the wake integrals: $s = 1 + y/y_0$, $dy = y_0\, ds$; $y \in [0, y_0]$ maps to $s \in [1, 2]$.

Force of the air on the airfoil (Notes Eq. 3.32), written on the wetted surface with the normal $\bar{n}$ of the control volume:

$$\boxed{\bar{F} = \int_{\Sigma_w} (p - p_\infty)\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

Hence the Σ_w terms of the momentum balance add up to $-\bar{F}$.

PHASE 3

Balances, Surface by Surface

`Mass (Notes Eq. 3.11)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot \bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-1)\,2y_{-\infty} = -2\rho U y_{-\infty}$
Σ_2	$2\rho\int_0^{y_0} \frac{U}{2}\left(1+\frac{y}{y_0}\right)dy = \frac{3}{2}\rho U y_0$
Σ_l	0 (streamlines, $\bar{v}\cdot\bar{n}=0$)
Σ_w	0 (impermeable wall)

$$-2\rho U y_{-\infty} + \frac{3}{2}\rho U y_0 = 0 \implies \boxed{y_{-\infty} = \frac{3}{4}\,y_0} \qquad (1)$$

The Σ_2 entry in detail: the antiderivative of $1 + y/y_0$ is $y + y^2/(2y_0)$, and by Barrow's rule between $y=0$ and $y=y_0$, $\rho U\left[y + \frac{y^2}{2y_0}\right]_0^{y_0} = \rho U\left(y_0 + \frac{y_0}{2}\right) = \frac{3}{2}\rho U y_0$ (the factor 2 of the table counts both halves of the wake).

`Linear momentum (Notes Eq. 3.36, with` $p_a = p_\infty$`)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_\infty)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_\infty)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-U)\,2y_{-\infty}\,\bar{e}_x=-2\rho U^2y_{-\infty}\,\bar{e}_x$	$0\ (p\simeq p_\infty)$	0 (uniform flow)
Σ_2	$2\rho\int_0^{y_0}v_x^2\,dy\,\bar{e}_x=\frac{7}{6}\rho U^2y_0\,\bar{e}_x$	$0\ (p\simeq p_\infty)$	0 (given)
Σ_l	$0\ (\bar{v}\cdot\bar{n}=0)$	$0\ (p\simeq p_\infty)$	0 (given)
Σ_w	0 **(impermeable)**	$-\int_{\Sigma_w}(p-p_\infty)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: the force of the airfoil on the air, $-\bar{F}$. The Σ_1 entry uses (1): $-2\rho U^2y_{-\infty}=-\frac{3}{2}\rho U^2y_0$.

$$-\frac{3}{2}\rho U^2y_0\,\bar{e}_x+\frac{7}{6}\rho U^2y_0\,\bar{e}_x=-\bar{F}\implies\boxed{\bar{F}=D\,\bar{e}_x,\qquad D=\frac{1}{3}\,\rho\,U^2\,y_0}\quad(2)$$

The Σ_2 momentum entry in detail, with $s=1+y/y_0$, $dy=y_0\,ds$ and Barrow's rule between $s=1$ and $s=2$: $2\rho\frac{U^2}{4}\int_0^{y_0}\left(1+\frac{y}{y_0}\right)^2dy=\frac{\rho U^2y_0}{2}\int_1^2s^2\,ds=\frac{\rho U^2y_0}{2}\left[\frac{s^3}{3}\right]_1^2=\frac{\rho U^2y_0}{2}\cdot\frac{8-1}{3}=\frac{7}{6}\rho U^2y_0$. The sum of the flux column is $\left(-\frac{9}{6}+\frac{7}{6}\right)\rho U^2y_0=-\frac{1}{3}\rho U^2y_0$.

For a wake symmetric about $y=0$ (a symmetric airfoil at zero incidence, as the symmetric profile implies) the y component of the momentum flux through Σ_2 cancels between the two halves, so $F_y=0$: no lift.

PHASE 4

Interpretation, Limits and Dimensional Check

- The wake carries less mass flux ($\frac{3}{2}\rho Uy_0$) than the undisturbed stream through the same width ($2\rho Uy_0$): the bounding streamlines are pushed outwards, from $y_{-\infty}=\frac{3}{4}y_0$ to y_0. The momentum deficit left in the wake is the drag.
- Limit $y_0\to0$: no wake, $y_{-\infty}\to0$ and $D\to0$ (d'Alembert's paradox for a frictionless attached flow).

- Dimensions (SI): $[y_{-\infty}] = \mathrm{m}$; $[D] = [\rho U^2 y_0] = \mathrm{N/m}$.
- Numerical cross-check (midpoint quadrature of the defining integrals with 2×10^5 cells, $U = 1$ m/s, $\rho = 1.2$ kg/m^3, $y_0 = 0.35$ m): $y_{-\infty}/y_0 = 0.75000$ and $D = 0.140000$ N/m $= \rho U^2 y_0/3$.
- Agreement with the official solution: CL1.pdf obtains $y_{-\infty} = \frac{3}{4} y_0$ and $\bar{F}_{\text{airfoil}} = \frac{1}{3}\rho U^2 y_0\, \bar{e}_x$.

Extension (beyond the official solution)

With the momentum thickness $\theta = \int_{-y_0}^{y_0} \frac{v_x}{U}\left(1 - \frac{v_x}{U}\right) dy = 2y_0 \int_0^1 \frac{1-\sigma^2}{4}\, d\sigma = \frac{y_0}{3}$ ($\sigma = y/y_0$), the result reads $D = \rho U^2 \theta$, the classical momentum-deficit formula for the drag from a wake survey.

PROBLEM CL2

Bore over a Stagnant Water Layer

Original statement · CL2.pdf

Moving frame | Mass and momentum | Hydrostatic pressure | Froude number

OFFICIAL PROBLEM STATEMENT

The bore sketched in the figure of CL2.pdf moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0, leaving behind a layer of thickness h_1 moving with velocity U_1. Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U, h_0 and g. As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.

Source: CL2.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Planar flow (per unit width) of an incompressible liquid of density ρ, with gravity $\bar{g} = -g\,\bar{e}_z$ (here gravity is **not** neglected: it drives the bore).
- The bore advances towards $-\bar{e}_x$ with constant velocity U over the stagnant layer h_0; behind it the layer h_1 moves with $-U_1\bar{e}_x$. Far from the front the velocity is uniform and horizontal and the free surface is flat at $p = p_a$.
- Viscous stresses negligible in the water sections and on the bottom wall (given); $\mu/(\rho U h) \ll 1$.
- Data: $[U,\ h_0,\ g,\ \rho]$. Unknowns: $[h_1,\ U_1]$ and the pressure distributions.

Symbol	Meaning	SI unit
U	velocity of the bore	m/s
U_1	velocity of the water behind the bore	m/s

Symbol	Meaning	SI unit
h_0, h_1	layer thickness ahead of and behind the bore	m
$F = U^2/(gh_0)$	Froude number of the bore	1

PHASE 2

Frames, Control Volume and Surfaces

- Laboratory frame L, with z measured from the bottom: unsteady flow, because the front moves.
- Frame of the bore B, moving with $-U\bar{e}_x$ relative to L. It is inertial (constant velocity), so the balances keep their form, and the flow is steady in it. Velocities: $\bar{v}_B = \bar{v}_L + U\bar{e}_x$, so the water ahead has $U\bar{e}_x$ and the water behind $(U - U_1)\bar{e}_x$.
- Control volume fixed in B ($\bar{v}_c = 0$). Surfaces: Σ_0, vertical section ahead ($\bar{n} = -\bar{e}_x$, height h_0); Σ_1, vertical section behind ($\bar{n} = +\bar{e}_x$, height h_1); Σ_b, bottom wall ($\bar{n} = -\bar{e}_z$); Σ_s, free surface (steady material surface in B, $p = p_a$).

PHASE 3

Balances, Surface by Surface

3.1 Preliminary: hydrostatic pressure in the uniform regions

Control volume: a column of width Δx in the uniform region ahead, from height z up to the free surface $z = h_0$. Surfaces: Σ_{bot} (at height z, $\bar{n} = -\bar{e}_z$), Σ_{top} (free surface, $\bar{n} = +\bar{e}_z$) and two vertical faces Σ_{side} ($\bar{n} = \mp\bar{e}_x$). Only the z component is needed.

> Momentum, z component, on the column (Notes Eq. 3.36)
>
> $$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho v_z \, dV}}_{0} + \int_{\Sigma_c} \rho v_z \, (\bar{v} \cdot \bar{n}) \, d\sigma = -\int_{\Sigma_c} p \, n_z \, d\sigma + \underbrace{\cancel{\int_{\Sigma_c} (\bar{\bar{\tau}}' \cdot \bar{n})_z \, d\sigma}}_{\text{negligible}} - \int_{V_c} \rho g \, dV$$
>
> Each surface integral, surface by surface (the vertical line drawn under each integral in class):
>
> | Surface | $\int \rho v_z \, (\bar{v} \cdot \bar{n}) \, d\sigma$ | $-\int p \, n_z \, d\sigma$ |
> |---|---|---|
> | Σ_{bot} | 0 ($v_z = 0$) | $+p(z)\,\Delta x$ |
> | Σ_{top} | 0 ($v_z = 0$) | $-p_a\,\Delta x$ |

Surface	$\int \rho v_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,n_z\,d\sigma$
Σ_{side}	0 ($v_z = 0$)	0 ($n_z = 0$)

Volume term: $-\int_{V_c} \rho g\,dV = -\rho g\,(h_0 - z)\,\Delta x$ (weight of the column).

$$0 = p(z)\,\Delta x - p_a\,\Delta x - \rho g\,(h_0 - z)\,\Delta x \implies \boxed{p = p_a + \rho g\,(h_0 - z)} \quad (1)$$

The same column taken behind the bore gives $p = p_a + \rho g\,(h_1 - z)$ (1').

3.2 Mass and momentum in the frame of the bore

`Mass (Notes Eq. 3.11), frame B`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_0	$\rho\,U\,(-1)\,h_0 = -\rho U h_0$
Σ_1	$\rho\,(U - U_1)\,h_1$
Σ_b	0 (wall)
Σ_s	0 (material surface)

$$\boxed{U\,h_0 = (U - U_1)\,h_1} \quad (2)$$

In the momentum balance p may be replaced by $p - p_a$ because $\int_{\Sigma_c} p_a \bar{n}\,d\sigma = 0$ on a closed surface. The x component is taken (gravity has no x component). On Σ_0 and Σ_1 the pressure is (1) and (1'); the antiderivative of $\rho g(h - z)$ is $\rho g(hz - z^2/2)$, and Barrow's rule between $z = 0$ and $z = h$ gives $\rho g h^2/2$.

Momentum, x component (Notes Eq. 3.36), frame B

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho v_x\,dV}_{0\ \text{(steady in } B)} + \int_{\Sigma_c}\rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,n_x\,d\sigma + \int_{\Sigma_c}(\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma + \underbrace{\int_{V_c}\rho g_x\,dV}_{g_x=0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,n_x\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$
Σ_0	$\rho U\,(-U)\,h_0 = -\rho U^2 h_0$	$+\int_0^{h_0}\rho g(h_0-z)\,dz = \frac{\rho g h_0^2}{2}$	0
Σ_1	$\rho\,(U-U_1)^2\,h_1$	$-\int_0^{h_1}\rho g(h_1-z)\,dz = -\frac{\rho g h_1^2}{2}$	0
Σ_b	0	0 ($n_x=0$)	0 (given)
Σ_s	0	0 ($p=p_a$)	0 (air)

$$\boxed{-\rho U^2 h_0 + \rho\,(U-U_1)^2\,h_1 = \frac{\rho g}{2}\left(h_0^2 - h_1^2\right)} \qquad (3)$$

3.3 Solution for h_1/h_0 and U_1/U

From (2), $U - U_1 = U h_0/h_1$, so $(U-U_1)^2 h_1 = U^2 h_0^2/h_1$. Substituting into (3) and dividing by ρ:

$$U^2 h_0\left(\frac{h_0}{h_1} - 1\right) = \frac{g}{2}(h_0 - h_1)(h_0 + h_1) \implies U^2 h_0\,\frac{h_0 - h_1}{h_1} = \frac{g}{2}(h_0 - h_1)(h_0 + h_1).$$

For a bore $h_1 \neq h_0$, so $h_0 - h_1$ cancels. With $r = h_1/h_0$ and $F = U^2/(g h_0)$, dividing by $g h_0^2$ gives $F/r = (1+r)/2$:

$$r^2 + r - 2F = 0 \implies \boxed{\frac{h_1}{h_0} = \frac{1}{2}\left(\sqrt{1+8F} - 1\right)} \qquad (4)$$

(the negative root is rejected, $h_1 > 0$). From (2), $U_1/U = 1 - h_0/h_1 = (r-1)/r$; with $s = \sqrt{1+8F}$, $r = (s-1)/2$ and

$$\boxed{\frac{U_1}{U} = 1 - \frac{2}{s-1} = \frac{\sqrt{1+8F}-3}{\sqrt{1+8F}-1}}, \qquad F = \frac{U^2}{g h_0}. \tag{5}$$

Erratum in the official solution (reading of the handwriting)

CL2.pdf gives $h_1/h_0 = \tfrac{1}{2}(\sqrt{1+8F}-1)$, as in (4). For the velocity it gives, as legible in the scan, $U_1/U = \tfrac{3}{2}\left(\tfrac{1}{3}\sqrt{1+8F}-1\right) = (\sqrt{1+8F}-3)/2 = r-1$. This contradicts its own continuity equation (2), which requires $U_1/U = (r-1)/r$; the two coincide only for $r = 1$. Check: (2) gives $(U-U_1)h_1 = U h_0 > 0$, hence $U_1 < U$, whereas $r - 1 > 1$ for $F > 3$ (for $F = 4$, $r - 1 = 1.3723$ against $U_1/U = 0.5785$ from (5)). The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- A bore exists only for $F > 1$: $r > 1 \iff s > 3 \iff F > 1$. F compares the speed of the front with the shallow-water wave speed $\sqrt{g h_0}$.
- Weak bore, $F = 1 + \epsilon$: $r \approx 1 + \tfrac{2}{3}\epsilon$ and $U_1/U \approx \tfrac{2}{3}\epsilon \to 0$ (an infinitesimal gravity wave travelling at $\sqrt{g h_0}$).
- Strong bore, $F \gg 1$: $r \approx \sqrt{2F}$ and $U_1 \to U$.
- Dimensions (SI): both sides of (3) are forces per unit width, $[\rho U^2 h_0] = [\rho g h^2] = \mathrm{N/m}$; F, r and U_1/U are dimensionless.
- Numerical cross-check: $F = 4$, $h_0 = 0.5$ m, $g = 9.81$ m/s^2 give $U = 4.4294$ m/s, $h_1/h_0 = 2.3723$ ($h_1 = 1.1861$ m), $U_1/U = 0.57846$ ($U_1 = 2.5623$ m/s); the residuals of (2) and (3) are below 10^{-14} in SI units.

Extension (beyond the official solution)

Mechanical energy is not conserved across the bore, so Bernoulli's equation cannot be used between both sections. The head lost is the classical hydraulic-jump value $\Delta H = (h_1 - h_0)^3/(4 h_0 h_1)$, positive for $h_1 > h_0$.

PROBLEM CL3

Double-Elbow Mouthpiece: Exit Velocity, Force and Moment

Original statement · CL3.pdf

Control volume | Linear momentum | Angular momentum | Gauge pressure

OFFICIAL PROBLEM STATEMENT

The double-elbow mouth piece of exit cross section A_o shown in the figure of CL3.pdf is to be installed at the end of a circular pipe of cross section A_p to redirect the outflow of water into the open atmosphere. Once the mouth piece is installed, the uniform values U_p and p_p are measured for the flow velocity and the pressure at the end of the pipe (i.e., just at the entrance of the mouth piece). Find the value of the uniform velocity at the exit section, U_o, as well as the force exerted on the mouth piece and its corresponding moment with respect to the entrance section. (Handwritten addition: neglect the effect of gravity.)

Source: CL3.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eqs. 3.37-3.38, force and moment on a body Eqs. 3.32-3.33.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible (water); neglect $\bar{g}$ (handwritten addition); uniform properties at inlet and outlet; viscous stresses negligible on both sections.
- Free jet at the exit: $p = p_a$ over Σ_o. The thin wall of the mouth piece has the atmosphere p_a on its outer face; the force asked for is the net force of the water inside plus the atmosphere outside, as in the official solution. Ignoring the atmosphere would add $p_a(A_p - A_o)\,\bar{e}_x$.
- Geometry (figure): S-shaped piece; the exit axis is a vertical distance H below the entrance axis; both the entrance and the exit flows are along $+\bar{e}_x$.
- Data: $[\rho,\ A_p,\ A_o,\ U_p,\ p_p,\ p_a,\ H]$. Unknowns: $[U_o,\ \bar{F}_{MP},\ \bar{M}_{MP}]$.

Symbol	Meaning	SI unit
A_p, A_o	entrance and exit areas	m^2

Symbol	Meaning	SI unit
U_p, p_p	velocity and pressure at the entrance	m/s, Pa
U_o	exit velocity	m/s
H	vertical offset between the axes	m
$\bar{F}_{MP}$, $\bar{M}_{MP}$	force and moment on the mouth piece	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the pipe: $\bar{e}_x$ along the pipe axis (flow direction), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Origin O at the centre of the entrance section (moments about O).
- Centre of the exit section: $\bar{x}_o = X\bar{e}_x - H\bar{e}_y$ relative to O (X drops out). Position within a circular section relative to its centre: $\bar{\eta}$, with $\int \bar{\eta}\, d\sigma = 0$.
- Control volume: the water inside the mouth piece, fixed ($\bar{v}_c = 0$). Surfaces: Σ_p (entrance, $\bar{n} = -\bar{e}_x$), Σ_o (exit, $\bar{n} = +\bar{e}_x$), Σ_w (inner wall).

Force and moment of the water and the atmosphere on the mouth piece, with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F}_{MP} = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

$$\boxed{\bar{M}_{MP} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_p	$\rho\,U_p\bar{e}_x\cdot(-\bar{e}_x)\,A_p=-\rho U_pA_p$
Σ_o	$\rho\,U_o\bar{e}_x\cdot\bar{e}_x\,A_o=+\rho U_oA_o$
Σ_w	0 (wall)

$$\rho U_pA_p=\rho U_oA_o\implies\boxed{U_o=\frac{A_p}{A_o}\,U_p},\qquad\dot{m}\equiv\rho U_pA_p=\rho U_oA_o\quad(1)$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_p	$\rho U_p\bar{e}_x\,(-U_p)\,A_p=-\dot{m}\,U_p\bar{e}_x$	$-(p_p-p_a)(-\bar{e}_x)A_p=(p_p-p_a)A_p\bar{e}_x$	0 (uniform)
Σ_o	$\rho U_o\bar{e}_x\,(U_o)\,A_o=\dot{m}\,U_o\bar{e}_x$	0 (free jet, $p=p_a$)	0 (uniform)
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}_{MP}$.

$$\dot{m}\,(U_o - U_p)\,\bar{e}_x = (p_p - p_a)A_p\,\bar{e}_x - \bar{F}_{MP} \implies \boxed{\bar{F}_{MP} = [(p_p - p_a)A_p + \rho U_p\,(U_p - U_o)\,A_p]\,\bar{e}_x} \quad (2)$$

No entry of the table has a component along $\bar{e}_y$ or $\bar{e}_z$, so $F_{MP,y} = F_{MP,z} = 0$.

Angular momentum about O (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\,\bar{x}\wedge\bar{v}\,dV}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\,\bar{x}\wedge\bar{g}\,dV}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_p	$-\rho U_p^2\left(\int \bar{\eta}\,d\sigma\right)\wedge\bar{e}_x = 0$	$(p_p-p_a)\left(\int \bar{\eta}\,d\sigma\right)\wedge\bar{e}_x = 0$	0
Σ_o	$\rho U_o^2 A_o\,(\bar{x}_o\wedge\bar{e}_x) = \rho U_o^2 H A_o\,\bar{e}_z$	$0\ (p = p_a)$	0
Σ_w	0	$-\int_{\Sigma_w} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int_{\Sigma_w} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed Σ_w terms: $-\bar{M}_{MP}$. In the Σ_o entry, $\bar{x} = \bar{x}_o + \bar{\eta}$ and $\bar{x}_o\wedge\bar{e}_x = (X\bar{e}_x - H\bar{e}_y)\wedge\bar{e}_x = -H\,\bar{e}_y\wedge\bar{e}_x = +H\,\bar{e}_z$.

$$\rho U_o^2 H A_o\,\bar{e}_z = -\bar{M}_{MP} \implies \boxed{\bar{M}_{MP} = -\rho U_o^2 H A_o\,\bar{e}_z = -\rho\,\frac{A_p^2}{A_o}\,U_p^2\,H\,\bar{e}_z} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $A_o < A_p$ gives $U_o > U_p$: the piece accelerates the water, and the wall supplies the momentum change.

- The moment exists only because the exit axis is offset by H: for $H \to 0$, $\bar{M}_{MP} \to 0$. With the exit below the entrance, the moment is clockwise as seen in the figure ($\bar{e}_z$ towards the reader).
- Limit $A_o \to A_p$: $U_o = U_p$ and $\bar{F}_{MP} \to (p_p - p_a) A_p \bar{e}_x$, the pure pressure force.
- Dimensions (SI): $[\rho U_p A_p (U_p - U_o)] = \mathrm{N}$; $[\rho U_o^2 H A_o] = \mathrm{N \cdot m}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $A_p = 0.02$ m^2, $A_o = 0.008$ m^2, $U_p = 3$ m/s, $p_p = 2.5 \times 10^5$ Pa, $p_a = 101325$ Pa, $H = 1.5$ m give $U_o = 7.5$ m/s, $F_{MP,x} = 2703.5$ N and $M_{MP,z} = -675$ N·m.
- Agreement with the official solution: CL3.pdf obtains (1), (2) and (3).

Extension (beyond the official solution)

If the flow inside were lossless (not stated), Bernoulli's equation $p_p - p_a = \frac{1}{2}\rho(U_o^2 - U_p^2)$ would give $F_{MP,x} = \frac{1}{2}\rho A_p (U_o - U_p)^2 \geq 0$. With losses, the measured p_p must be used, as in (2).

PROBLEM CL4

Cascade of Aerofoils: Downstream State

Original statement · CL4.pdf

Periodicity | Mass, momentum, energy | Perfect gas | Cascade

OFFICIAL PROBLEM STATEMENT

A planar uniform stream of air with pressure p_1, density ρ_1 and velocity $\bar{v}_1 = v_{1x}\bar{e}_x$ flows past an array of aerofoils with vertical separation L, exerting on each one a force (per unit length) given by $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2, density ρ_2 and velocity $\bar{v}_2 = v_{2x}\bar{e}_x + v_{2y}\bar{e}_y$ for the uniform stream found sufficiently far downstream.

Source: CL4.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar flow through an infinite array of identical aerofoils (pitch L), so the flow is periodic in y; uniform far upstream (1) and far downstream (2).
- Neglect $\bar{g}$; aerofoils thermally insulated; compressible fluid. Viscous stresses and heat conduction negligible in the uniform sections.
- Perfect gas: $h = e + \frac{p}{\rho} = c_pT = \frac{c_p}{R_g}\frac{p}{\rho} = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$ (with $p = \rho R_g T$ and $\gamma = c_p/c_v$), as in the class solution.
- Data: $[p_1,\ \rho_1,\ v_{1x},\ L,\ F_x,\ F_y,\ \gamma]$. Unknowns: $[p_2,\ \rho_2,\ v_{2x},\ v_{2y}]$: four equations are needed.

Symbol	Meaning	SI unit
p_1, ρ_1, v_{1x}	upstream state	Pa, kg/m^3, m/s
$p_2, \rho_2, v_{2x}, v_{2y}$	downstream state	Pa, kg/m^3, m/s
L	pitch of the array	m

Symbol	Meaning	SI unit
F_x, F_y	force on each aerofoil per unit length	N/m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the array; $\bar{e}_x$ along the incoming stream, $\bar{e}_y$ along the array. Fixed control volume, $\bar{v}_c = 0$, one period high.
- Surfaces: Σ_1 (far upstream, height L, $\bar{n} = -\bar{e}_x$), Σ_2 (far downstream, height L, $\bar{n} = +\bar{e}_x$), Σ_{l1} and Σ_{l2} (upper and lower stream surfaces, $\bar{v} \cdot \bar{n} = 0$), Σ_w (aerofoil surface, $\bar{v} = 0$, insulated).
- Periodicity: Σ_{l2} is Σ_{l1} shifted by $-L\bar{e}_y$ with opposite normals, so the integrals of any periodic quantity over Σ_{l1} and Σ_{l2} cancel each other out.

Force on each aerofoil (per unit length), with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y = \int_{\Sigma_w} p\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v} \cdot \bar{n}\,d\sigma$
Σ_1	$\rho_1(-v_{1x})\,L$

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_2	$\rho_2\,v_{2x}\,L$
$\Sigma_{l1},\ \Sigma_{l2}$	0 (stream surfaces)
Σ_w	0 (wall)

$$\boxed{\rho_1 v_{1x} = \rho_2 v_{2x}} \qquad (1)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1 v_{1x}\bar{e}_x(-v_{1x})L$	$+p_1 L\,\bar{e}_x$	0
Σ_2	$\rho_2(v_{2x}\bar{e}_x + v_{2y}\bar{e}_y)\,v_{2x}L$	$-p_2 L\,\bar{e}_x$	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	$-\int_{\Sigma_w} p\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F} = -F_x\bar{e}_x - F_y\bar{e}_y$. With (1), $\rho_2 v_{2x}L = \rho_1 v_{1x}L$ in the Σ_2 flux.

$$\boxed{F_x = \rho_1 v_{1x} L\,(v_{1x}-v_{2x}) + (p_1-p_2)\,L} \quad (2) \qquad \boxed{F_y = -\rho_1 v_{1x} L\,v_{2y}} \quad (3)$$

`Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\tfrac{v^2}{2}\right)dV}_{0} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}(Q_c+Q_r)\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1\left(e_1+\frac{v_{1x}^2}{2}\right)(-v_{1x})L$	$+p_1 v_{1x} L$	0	0
Σ_2	$\rho_2\left(e_2+\frac{v_{2x}^2+v_{2y}^2}{2}\right)v_{2x}L$	$-p_2 v_{2x} L$	0	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	0 ($\bar{v}=0$)	0 ($\bar{v}=0$)	0 (insulated)

Moving the pressure work to the left turns e into $h=e+p/\rho$; dividing by $\rho_1 v_{1x} L = \rho_2 v_{2x} L$ and using $h=\frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\frac{\gamma}{\gamma-1}\left(\frac{p_2}{\rho_2}-\frac{p_1}{\rho_1}\right)+\frac{v_{2x}^2+v_{2y}^2}{2}-\frac{v_{1x}^2}{2}=0} \quad (4)$$

(1) to (4) are four equations for the four unknowns p_2, ρ_2, v_{2x}, v_{2y}, which answers the question.

PHASE 4

Interpretation, Limits and Dimensional Check

- (3): the transverse force on an aerofoil equals minus the y momentum given to the stream; the flow is deflected opposite to the force. (2) splits the streamwise force into a pressure drop and a momentum change.
- $\bar{F}=0$: (1) to (4) are satisfied by state 2 = state 1. Incompressible limit ($\rho_2=\rho_1$): $v_{2x}=v_{1x}$ and $p_1-p_2=F_x/L$.
- Dimensions (SI): $[F_x/L]=\mathrm{Pa}=[\rho_1 v_{1x}^2]$; $[\rho_1 v_{1x} L v_{2y}]=\mathrm{N/m}$; $[\gamma p/((\gamma-1)\rho)]=\mathrm{m^2/s^2}=[v^2]$.
- Agreement with the official solution and the class solution: the same four equations (1) to (4).

Extension (beyond the official solution): closed solution

With $m = \rho_1 v_{1x}$: $\rho_2 = m/v_{2x}$, $v_{2y} = -F_y/(mL)$ and $p_2 = P - m\,v_{2x}$ with $P = p_1 + mv_{1x} - F_x/L$. Substituting into (4), with $E = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_{1x}^2}{2}$ and the coefficient $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$ of v_{2x}^2: $(\gamma+1)v_{2x}^2 - 2\gamma\frac{P}{m}v_{2x} + 2(\gamma-1)\left(E - \frac{v_{2y}^2}{2}\right) = 0$. For $\bar{F} = 0$ one root is $v_{2x} = v_{1x}$; the root continuous with it is the physical one. Check: $p_1 = 101325$ Pa, $\rho_1 = 1.2$ kg/m^3, $v_{1x} = 100$ m/s, $L = 0.2$ m, $F_x = 30$ N/m, $F_y = -80$ N/m, $\gamma = 1.4$ give $v_{2x} = 100.16$ m/s, $v_{2y} = 3.3333$ m/s, $\rho_2 = 1.19808$ kg/m^3, $p_2 = 101155.8$ Pa, residuals of (1) to (4) below 10^{-9}; the other root gives $p_2 = -6843.3$ Pa < 0 and is discarded.

PROBLEM CL5

Turbojet on a Testbench: Exit State

Original statement · CL5.pdf

Moving boundaries | Mass, momentum, energy | Heat and work | Thrust

OFFICIAL PROBLEM STATEMENT

The entrance and exit sections of the turboreactor sketched in the figure of CL5.pdf have areas A_1 and A_2. Besides the outer fixed frame, the machine has mobile parts in its interior that extract an amount of work W per unit time from the fluid stream. The combined effects of chemical reaction in the combustion chamber and heat transfer through the walls give a heat transfer per unit time Q to the fluid stream. The axial force on the turboreactor measured on the testbench is $\bar{F} = -F\,\bar{e}_x$. For given values of W, Q and F and assuming that at the entrance and exit sections the flow properties are uniform, obtain the values of the density ρ_2, pressure p_2 and velocity v_2 in terms of their values p_1, ρ_1 and v_1 at the entrance. (Handwritten addition: no gravity.)

Source: CL5.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Compressible flow; neglect gravity; uniform properties at inlet and outlet; steady on average (the rotating parts make the flow periodic inside, but there is no net accumulation).
- Perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$, as in the class solution. Fuel mass flow neglected (not given), so the same mass flow crosses both sections.
- $\dot{W}$: work per unit time extracted from the fluid by the mobile parts; $\dot{Q}$: heat per unit time transferred to the fluid; $\bar{F} = -F\bar{e}_x$: force on the device measured on the testbench (fluid inside plus ambient air outside, at p_a).
- Data: $[\rho_1,\ p_1,\ \bar{v}_1 = v_1\bar{e}_x,\ A_1,\ A_2,\ p_a,\ F,\ \dot{W},\ \dot{Q},\ \gamma]$. Unknowns: $[\rho_2,\ p_2,\ \bar{v}_2 = v_2\bar{e}_x]$.

Symbol	Meaning	SI unit
A_1, A_2	entrance and exit areas	m^2

Symbol	Meaning	SI unit
p_1, ρ_1, v_1	entrance state	Pa, kg/m^3, m/s
p_2, ρ_2, v_2	exit state	Pa, kg/m^3, m/s
F	thrust measured on the bench	N
$\dot{W}, \dot{Q}$	power extracted and heat added	W
$\dot{m} = \rho_1 v_1 A_1$	mass flow	kg/s

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the testbench; $\bar{e}_x$ along the engine axis in the flow direction.
- Surfaces: Σ_1 (entrance, $\bar{n} = -\bar{e}_x$), Σ_2 (exit, $\bar{n} = +\bar{e}_x$), Σ_w (fixed walls, $\bar{v} = \bar{v}_c = 0$) and Σ_m (surface of the mobile parts, moving with them: $\bar{v} = \bar{v}_c \neq 0$). On Σ_w and Σ_m, $(\bar{v} - \bar{v}_c) \cdot \bar{n} = 0$.
- $\bar{n}$ is the outward normal of the control volume; $\bar{n}_{\text{in}} = -\bar{n}$ points into the fluid.

Definitions written on the surfaces, as in class:

$$\boxed{\bar{F} = \int_{\Sigma_m, \Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_m, \Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma = -F\, \bar{e}_x}$$

$$\boxed{\dot{Q} = \int_{\Sigma_w, \Sigma_m} \bar{q} \cdot \bar{n}_{\text{in}}\, d\sigma + \int_{V_c} Q_R\, dV}$$

$$\boxed{\dot{W} = -\int_{\Sigma_m} p\, \bar{n}_{\text{in}} \cdot \bar{v}\, d\sigma + \int_{\Sigma_m} (\bar{\bar{\tau}}' \cdot \bar{n}_{\text{in}}) \cdot \bar{v}\, d\sigma}$$

Only Σ_m enters $\dot{W}$: Σ_w is a fixed surface and does no work. $\dot{W}$ is the power of the gas on the moving solid, so the power done on the gas there is $-\dot{W}$.

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.10)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\text{ (on average)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho_1 v_1 A_1$
Σ_2	$+\rho_2 v_2 A_2$
Σ_m	$0\ (\bar{v}=\bar{v}_c)$
Σ_w	0

$$\boxed{\rho_1 v_1 A_1=\rho_2 v_2 A_2=\dot{m}} \qquad (1)$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1 v_1\bar{e}_x(-v_1)A_1=-\dot{m}v_1\bar{e}_x$	$(p_1-p_a)A_1\bar{e}_x$	0
Σ_2	$\rho_2 v_2\bar{e}_x(v_2)A_2=\dot{m}v_2\bar{e}_x$	$-(p_2-p_a)A_2\bar{e}_x$	0

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
$\Sigma_m,\ \Sigma_w$	0	$-\int_{\Sigma_m,\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_m,\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_m, Σ_w terms: $-\bar{F} = F\bar{e}_x$ (uniform velocity vectors give no viscous stress on Σ_1, Σ_2).

$$\boxed{\dot{m}\,(v_2 - v_1) = (p_1 - p_a)A_1 - (p_2 - p_a)A_2 + F} \tag{2}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\left(e+\tfrac{v^2}{2}\right)dV}}_{0} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}}_{\text{neglect}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int\bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\dot{m}\left(e_1+\frac{v_1^2}{2}\right)$	$+p_1v_1A_1$	0	$0\ (\nabla T=0)$
Σ_2	$+\dot{m}\left(e_2+\frac{v_2^2}{2}\right)$	$-p_2v_2A_2$	0	$0\ (\nabla T=0)$
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m}(\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int_{\Sigma_m}\bar{q}\cdot\bar{n}\,d\sigma$
Σ_w	0	$0\ (\bar{v}=0)$	$0\ (\bar{v}=0)$	$-\int_{\Sigma_w}\bar{q}\cdot\bar{n}\,d\sigma$

Boxed terms: the pressure and viscous work on Σ_m add up to $-\dot{W}$; the conduction through Σ_m, Σ_w plus the volume term $\int_{V_c} Q_R\,dV$ add up to $\dot{Q}$. Moving the pressure work on Σ_1, Σ_2 to the left gives $h = e + p/\rho$; with (1) and $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\dot{m}\left[\left(\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{v_2^2}{2}\right)-\left(\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{v_1^2}{2}\right)\right] = -\dot{W}+\dot{Q}} \tag{3}$$

(1), (2) and (3) determine ρ_2, p_2 and v_2.

PHASE 4

Interpretation, Limits and Dimensional Check

- (2) is the thrust balance: thrust = change of momentum flux + pressure imbalance on the sections. For $p_1 = p_2 = p_a$, $F = \dot{m}(v_2 - v_1)$.
- (3): the stagnation enthalpy of the stream rises by the heat added and falls by the work extracted, per unit mass flow.
- Dimensions (SI): $[\dot{m}(v_2 - v_1)] = \mathrm{N}$; $[(\dot{Q} - \dot{W})/\dot{m}] = \mathrm{J/kg} = \mathrm{m^2/s^2} = [h] = [v^2]$.
- Numerical illustration (hypothetical data): $A_1 = 0.5\ \mathrm{m^2}$, $A_2 = 0.3\ \mathrm{m^2}$, $v_1 = 150$ m/s, $p_1 = 0.9 \times 10^5$ Pa, $\rho_1 = 1.1$ kg/m^3 ($M_1 = 0.443$), $p_a = 10^5$ Pa, $F = 30$ kN, $\dot{W} = 2$ MW, $\dot{Q} = 30$ MW, $\gamma = 1.4$: $\dot{m} = 82.5$ kg/s and the two roots of the extension below, $v_2 = 355.515$ m/s, $p_2 = 126817$ Pa, $\rho_2 = 0.77353$ kg/m^3 ($M_2 = 0.742$), and $v_2 = 597.263$ m/s, $p_2 = 60336$ Pa, $\rho_2 = 0.46043$ kg/m^3 ($M_2 = 1.39$); residuals of (1) to (3) below 10^{-6} in SI units.
- Agreement with the official and class solutions: the same equations (1), (2), (3) (CL5.pdf writes (2) as $F = \rho_1 v_1 A_1(v_2 - v_1) - (p_1 - p_a)A_1 + (p_2 - p_a)A_2$).

Extension (beyond the official solution): explicit solution

With $K = p_a A_2 + F + (p_1 - p_a)A_1 + \dot{m}v_1$, (2) gives $p_2 A_2 = K - \dot{m}v_2$ and, with (1), $p_2/\rho_2 = (K - \dot{m}v_2)v_2/\dot{m}$. Substituting into (3) and using $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$: $(\gamma+1)v_2^2 - 2\gamma\frac{K}{\dot{m}}v_2 + 2(\gamma-1)H_0 = 0$, with $H_0 = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_1^2}{2} + \frac{\dot{Q}-\dot{W}}{\dot{m}}$. The two roots are a subsonic and a supersonic exit state; for a subsonic entrance with heat addition the physical one is the subsonic (smaller) root, since reaching the other would require passing through sonic conditions inside the engine, which this overall balance cannot describe. For $F = 0$, $\dot{Q} = \dot{W} = 0$ and $A_1 = A_2$ one root is $v_2 = v_1$ and the product of the roots is $2(\gamma-1)H_0/(\gamma+1)$, the Prandtl relation of a normal shock.